



Image Data Types

There is a wide variety of medical images. Because of this medical images are stored in a variety of ways. When working with medical images, you'll find it useful to have knowledge of the various data types used to store images so you can work with them effectively.

Navigate to the Module 1 folder in the course files and work through the **ImageDataTypes.mlx** script. You'll work with the X-ray image you saw in the previous video and learn how to convert between different data types.

While we recommend you run and experiment with the code in MALTAB, you can read a static version below. The interactive tools are not present in the static version.

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Converting to Different Data Types

Depending on the modality and specific equipment used, your images may be stored in a variety of data types. Your analysis may require converting data types to make all your images consistent. Additionally, it is best to convert your images to double precision when performing arithmetic operations on your data to have the highest precision and avoid errors.

There are several functions to help you convert data types.

- `im2uint8` - convert to 8-bit unsigned integers (0 - 255)
- `im2uint16` - convert to 16-bit unsigned integers (0 - 65535)
- `im2int16` - convert to 16-bit signed integers (-32768 - 32767)
- `im2single` - convert to single precision (32-bit) and rescales values to the range [0, 1]
- `im2double` - converts to double precision (64-bit) and rescales values to the range [0, 1]

Both the original image and the contrast-adjusted image have data type `uint16`. The code below converts both images to double precision and displays them. Notice that the range of values is now [0, 1]. However, the conversion maintains the relative intensity before and after the conversion. So, the original data converted to double precision still appears dark.

```
imgd = im2double(img);
imgadj = im2double(imgadj);

% Create a tiled layout to view the images side-by-side
figure
tiledlayout(1,2)
nexttile

imshow(imgd)
title("Original data converted to double")
colorbar

nexttile
imshow(imgadj)
title("Adjusted data converted to double")
colorbar
```

Further Resources

There are many possible contrast adjustments and operations you might perform on your images. Once you have your data in MATLAB and properly scaled, you work with medical images just like other images. If you haven't already, enroll [Introduction to Image Processing](#) to learn more about common image operations.

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